

THE WEEK IN ONE MODEL: FOUNDATION MODELS

Lecturer: Dr. Ting-Yun (Sunny) Cheng

Bridge of the week

02

Foundation models

03

Foundation models for Astronomy

04

Discussion

05

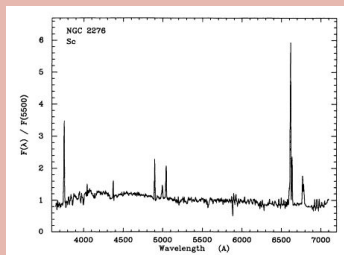
WHAT IS MACHINE LEARNING?

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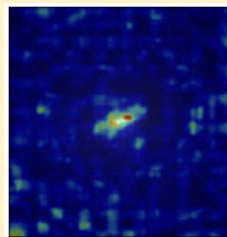
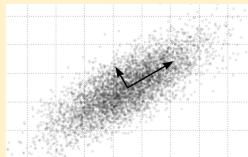
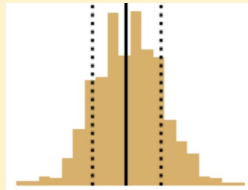
*Systems that learn patterns from data to make predictions/ decisions **without** being explicitly programmed with fixed rules (e.g. theoretical models).*

FROM DATA TO PATTERN

Raw Data



Feature Extraction



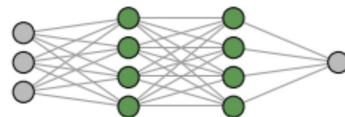
Find Pattern

$$\hat{y} = f(\mathbf{x}, \theta)$$

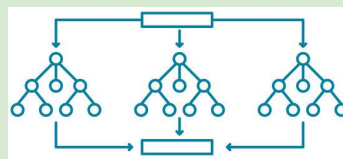
Linear model

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b$$

Neural Networks

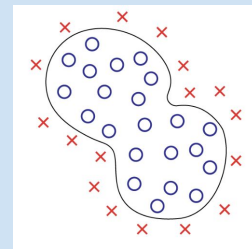
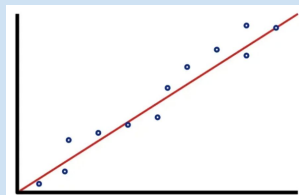
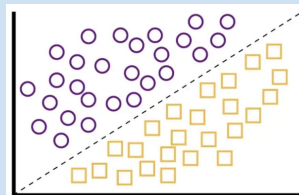


Decision Tree



Others

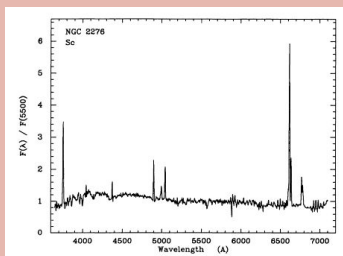
Prediction



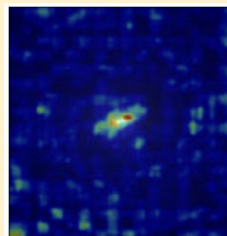
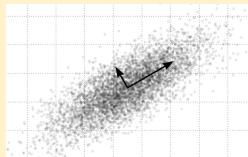
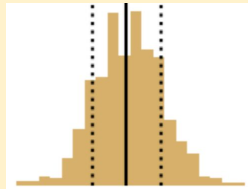
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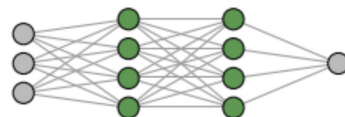
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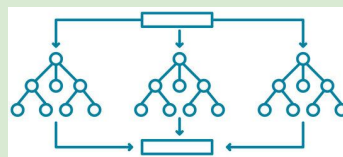
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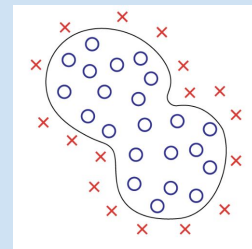
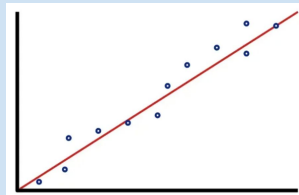
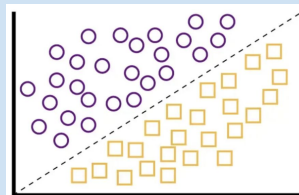


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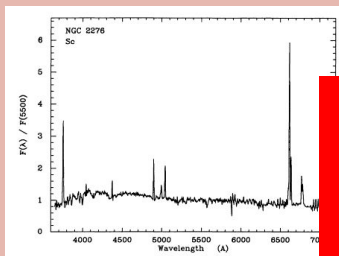
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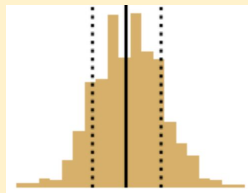


FROM DATA TO PATTERN

Raw Data



Feature Extraction



What foundation models can or try to expand to.

Find Pattern

$$\hat{y} = f(\mathbf{x}, \theta)$$

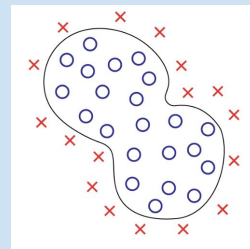
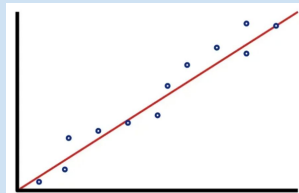
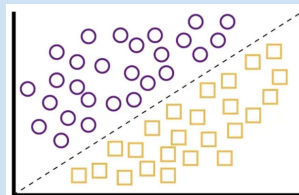
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VISTULA

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FOUNDATION MODELS

FROM ONE MODEL PER TASK TO ONE MODEL FOR ALL

By definition ([Stanford 2021](#)):

- Massive Scale (Pre-training)
- Self-Supervised Learning
- Homogenization & Adaptability

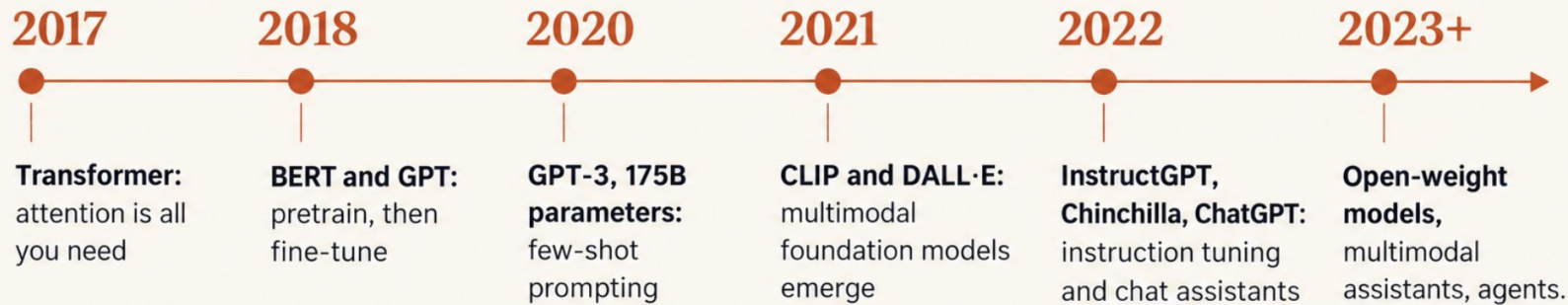
FOUNDATION MODELS

FROM ONE MODEL PER TASK TO ONE MODEL FOR ALL

- **Language models (LLMs):** on text to understand and generate language (e.g., GPT, Claude)
- **Vision models:** on images for recognition, segmentation, or generation (e.g., ViT, Stable Diffusion)
- **Multimodal models:** handle several data types together, such as text, images, and audio (e.g., CLIP, Gemini, GPT-4o)
- **Audio and speech models:** transcribe, translate, or generate speech and sound (e.g., Whisper).
- **Scientific and domain models:** built for specific fields like biology, chemistry (e.g., AlphaFold [Google], ESM [Meta])

FOUNDATION MODELS

SIX YEARS THAT CHANGE THE FIELD AND BEYOND



- [Attention Is All You Need](#) (Vaswani+17; Google)
- [BERT \(Bidirectional Encoder Representations from Transformers\)](#) (Devlin+18; Google)
- [GPT foundation paper](#) (Radford+18; OpenAI)
- [CLIP model](#) (Radford+21; OpenAI), [DALL-E model](#) (Ramesh+21; OpenAI)
- GPT model first came online in 2022

FOUNDATION MODELS

HOW TO BUILD A FOUNDATION MODEL?

A few key ingredients (and keywords):

- Tokenisation and Embeddings
- Transformer
- Self-supervised learning

TOKENISATION

LLM DO NOT SEE WORDS – THEY READ **TOKEN**

Early Tokenization Techniques & Limitations

Word-Level Tokenization

The quick brown fox.

Massive
Vocabulary!



Unseen Words?



Character-Level Tokenization

T h e q u i c k

Excessively Long Sequences



High Computational Cost

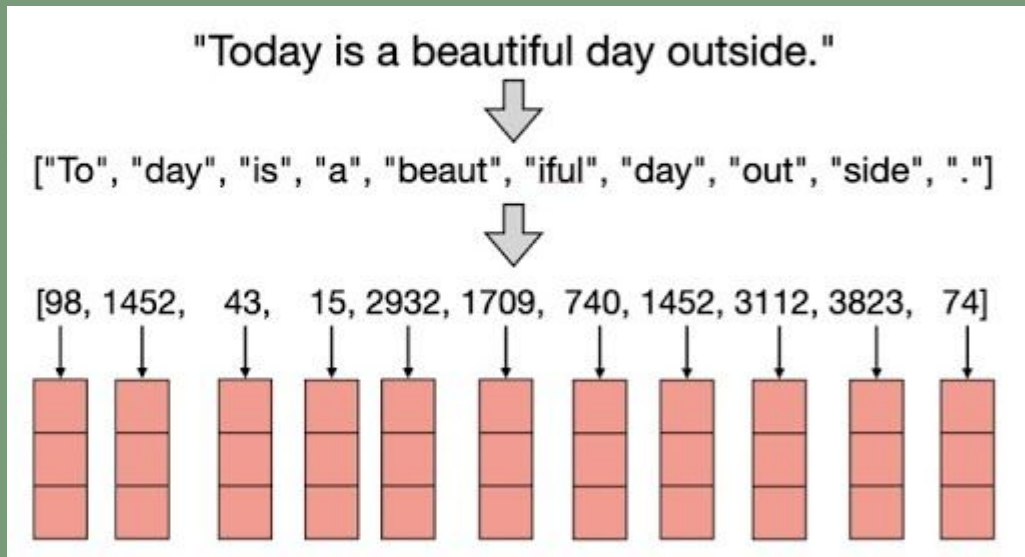


TOKENISATION

<https://platform.openai.com/tokenizer>

LLM DO NOT SEE WORDS – THEY READ **TOKEN**

Byte-pair encoding (first introduced in 1994 by Phillip Gage):
replace the most frequent pairs of bytes with a new byte



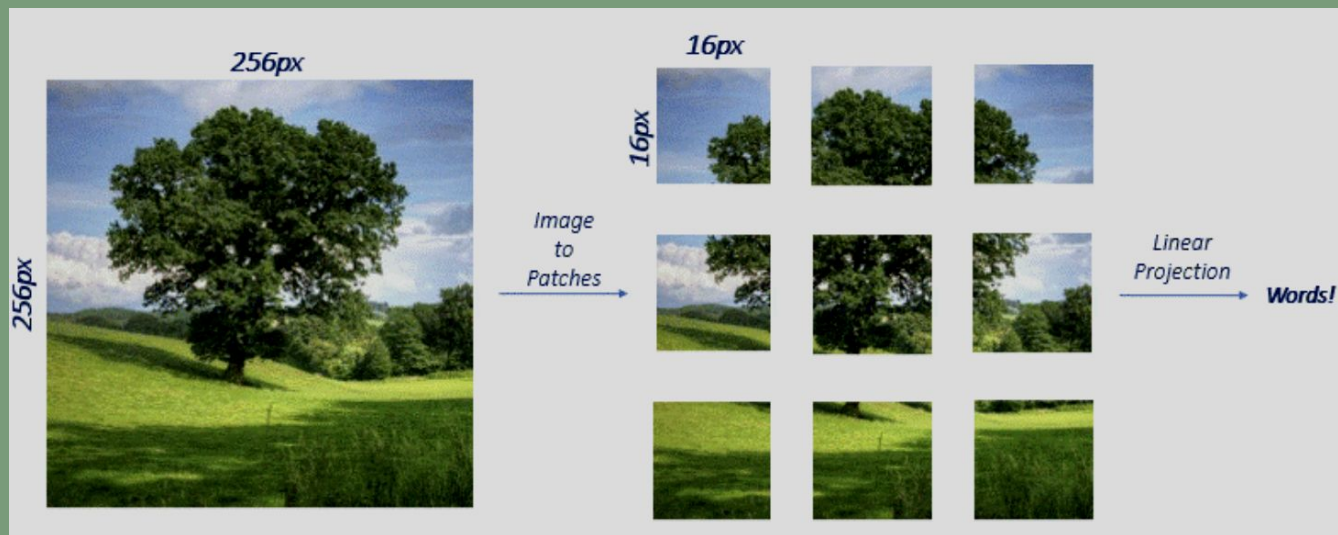
- **GPT-3.5:** 4,096 tokens (~3,000 words)
- **GPT-4 (Original):** 8,192 to 32,768 tokens (~6,000 to 24,000 words)
- **GPT-4o / GPT-4 Turbo:** 128,000 tokens (~96,000 words)

TOKENISATION FOR DIFFERENT DATA TYPE

Image Tokenisation: [Dosovitskiy+21](#)

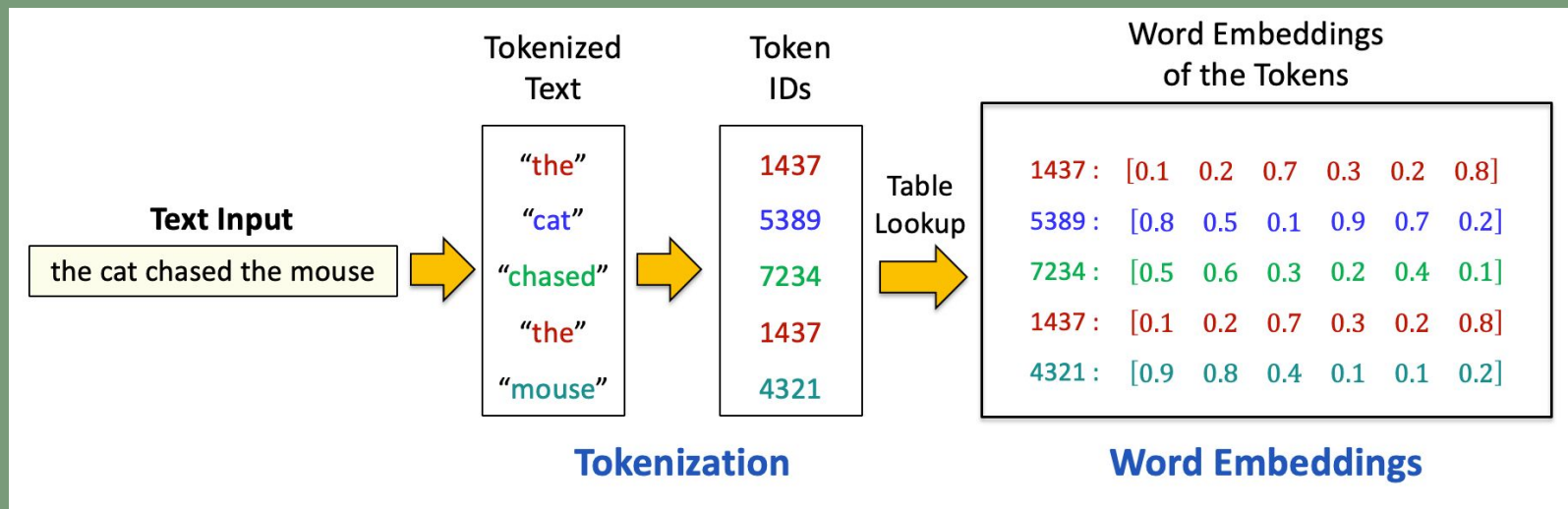
An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby



EMBEDDINGS

Embeddings are vectors stored in a lookup table, where each token is assigned a position in a high-dimensional space.



EMBEDDINGS

However... context and surrounding words are important

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It is shit

It is the shit

EMBEDDINGS

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It is shit → It is terrible

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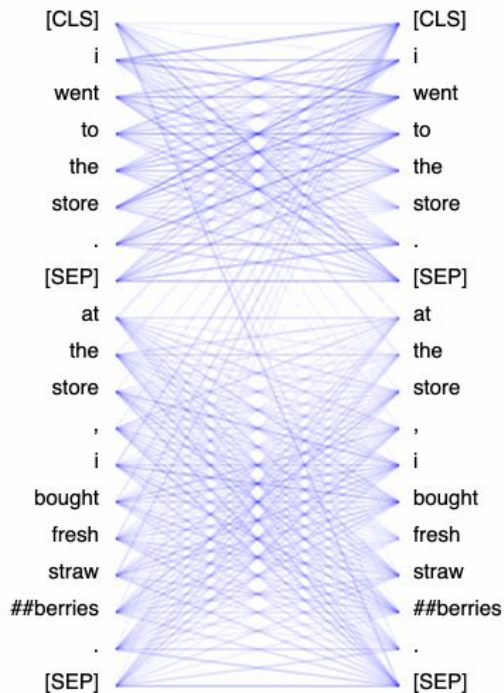
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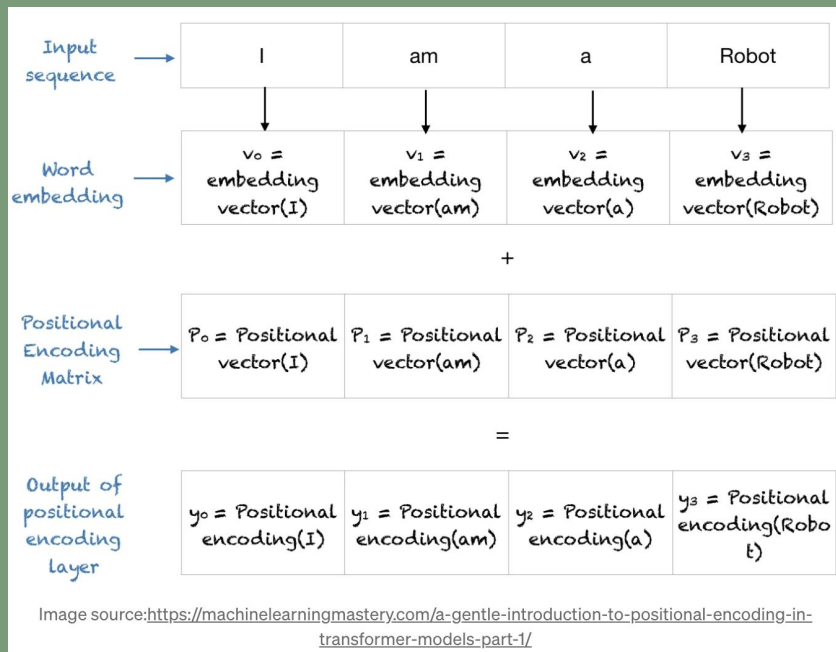
TRANSFORMER: ATTENTION IS ALL YOU NEED

<https://arxiv.org/abs/1706.03762>



TRANSFORMER: QUERY, KEY AND VALUE

Content embeddings (from tokens): $x_1, x_2, \dots, x_i$
 Position embeddings: $p_1, p_2, \dots, p_i$ $\rightarrow h_i = x_i + p_i$



TRANSFORMER: QUERY, KEY AND VALUE

Content embeddings (from tokens): $x_1, x_2, \dots, x_i$ $\rightarrow h_i = x_i + p_i$
 Position embeddings: $p_1, p_2, \dots, p_i$

Query (Q): "What information am I looking for?"

Key (K): "What information do I contain?"

Value (V): "Here is the information you can actually use."

$$q_i = h_i W_Q$$

$$k_i = h_i W_K$$

$$v_i = h_i W_V$$

$$A_{ij} = \frac{\exp(q_i k_j^T / \sqrt{d_k})}{\sum_j \exp(q_i k_j^T / \sqrt{d_k})}$$

$$z_i = \sum_j A_{ij} v_j$$

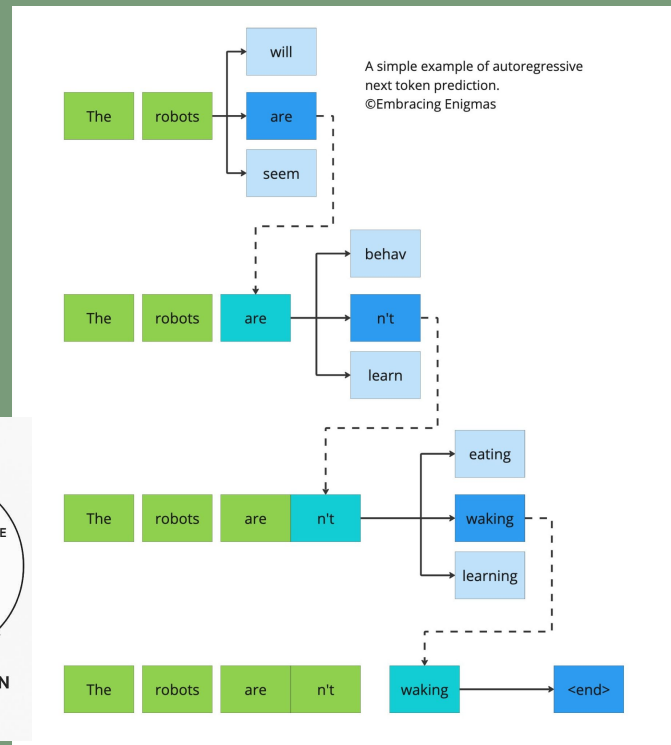
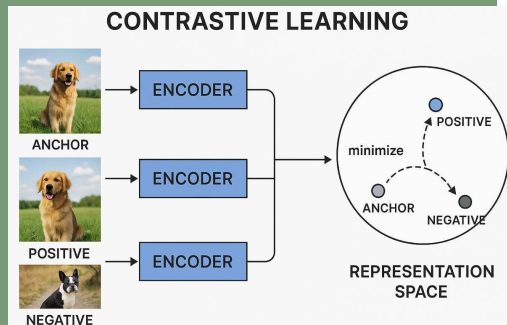
SELF-SUPERVISED LEARNING

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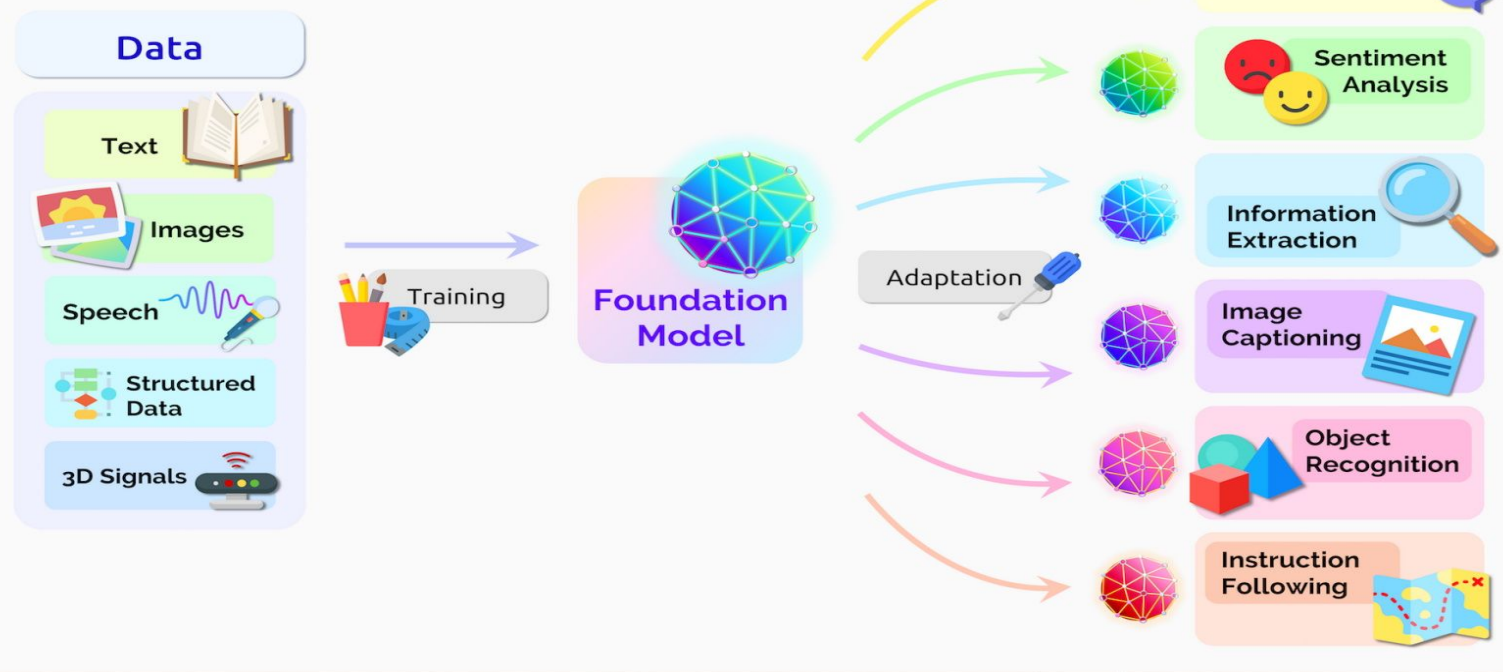
- Next-token prediction (autoregressive): GPT, Claude
- Masked prediction: BERT
- Contrastive learning: CLIP



FOUNDATION MODEL IN ONE PLOT

Source: NVIDIA blog –

<https://blogs.nvidia.com/blog/what-are-foundation-models/>



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FOUNDATION MODELS IN ASTRONOMY

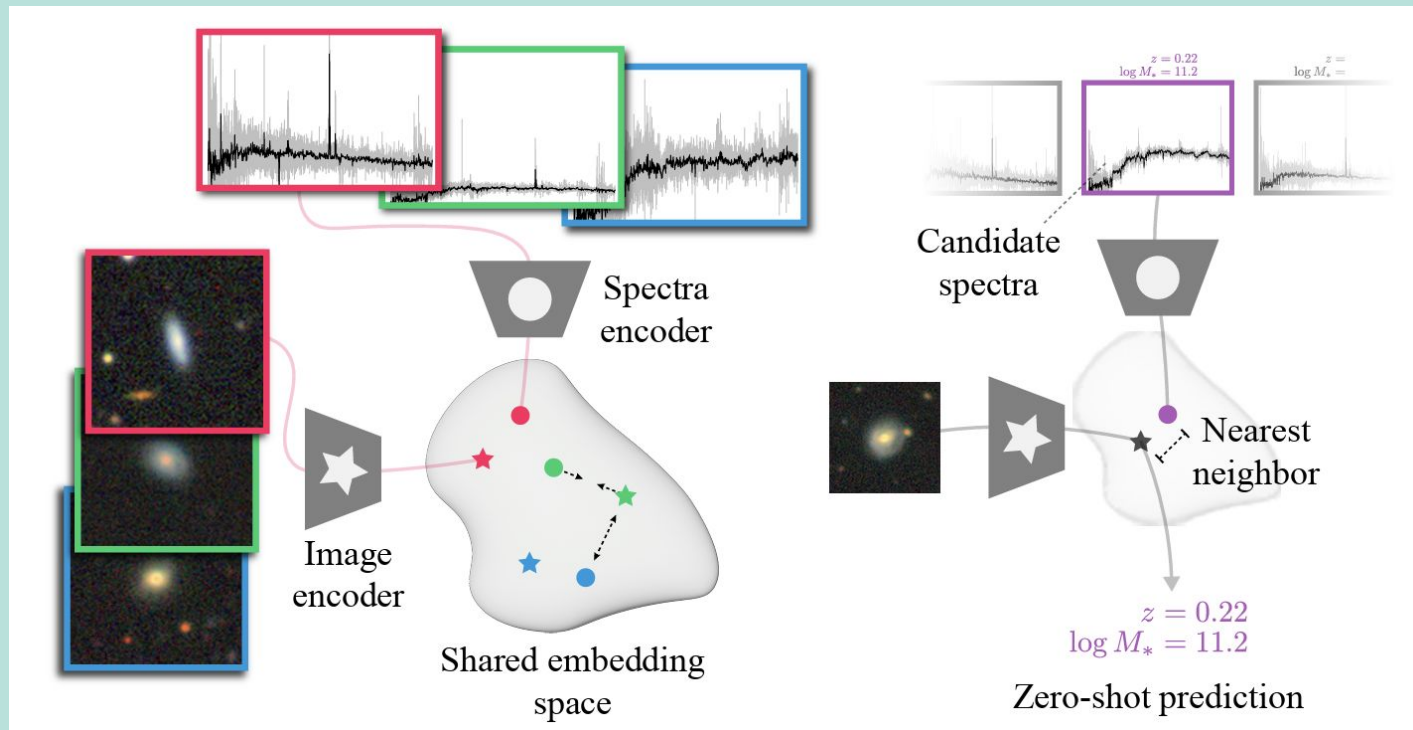
- Datasets: [Multimodal Universe](#) - Enabling Large-Scale Machine Learning with 100TBs of Astronomical Scientific Data



Survey	Modality	Science Use Case	# samples
Legacy Surveys DR10	Images	Galaxies	124M
Legacy Surveys North	Images	Galaxies	15M
Gaia BP/RP	Spectra	Stars	220M
SDSS-II	Spectra	Galaxies, Stars	4M
DESI	Spectra	Galaxies	20M
MaNGA SDSS-IV	Hyperspectral Image	Galaxies	12k
PLAsTiCC	Time Series	Time-varying objects	3.5M
TESS	Time Series	Exoplanets	160k

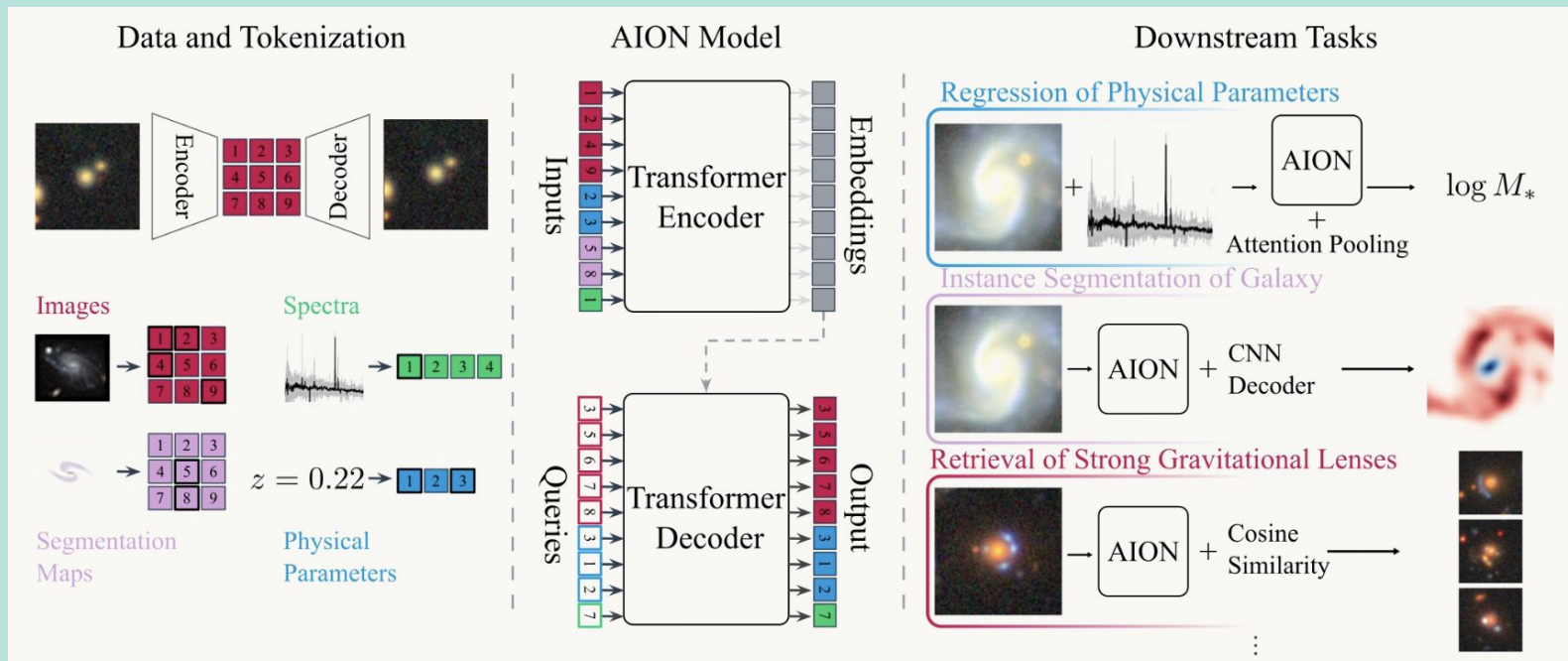
FOUNDATION MODELS IN ASTRONOMY

- [AstroCLIP](#) (Parker+23)



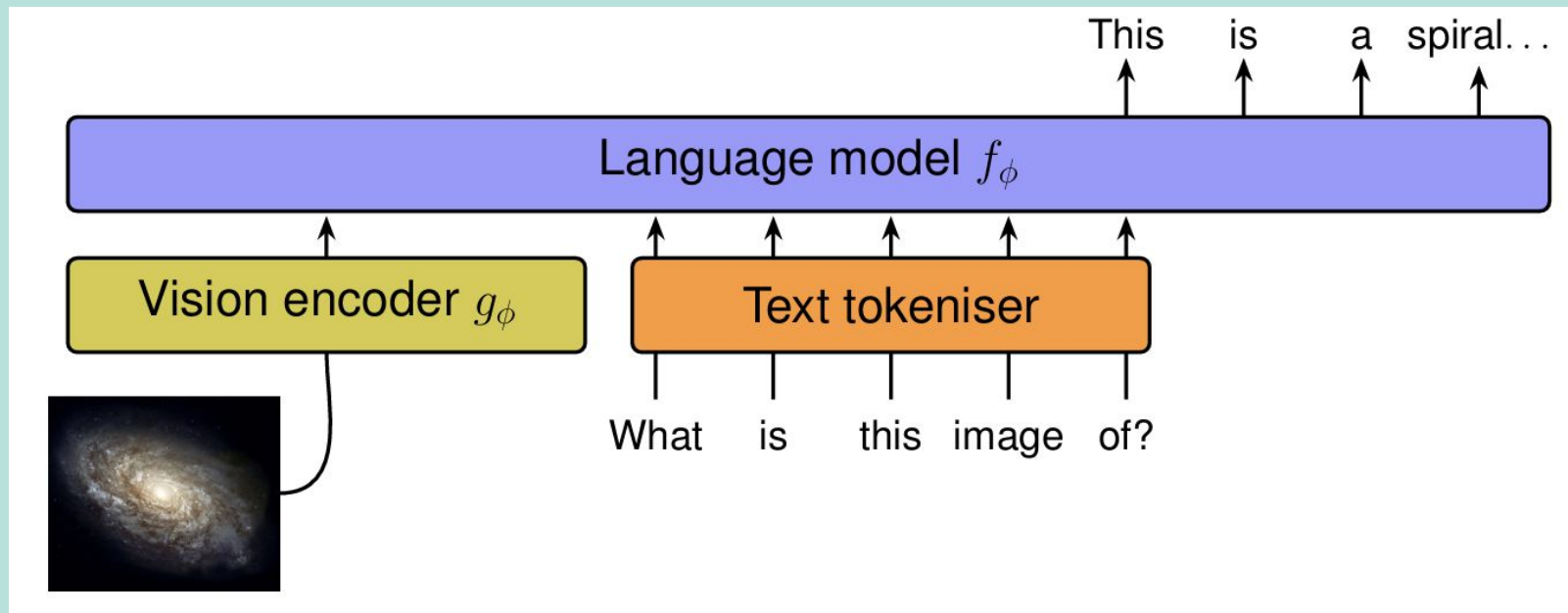
FOUNDATION MODELS IN ASTRONOMY

- AION-1 (Parker+25)



FOUNDATION MODELS IN ASTRONOMY

- [AstroLLaVA](#) (Saman+25): towards the unification of astronomical data and natural language



DISCUSSION TO THE END...

DO WE NEED COMPLEX MODELS?

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My opinion: often we don't.

A complex model is worth it only when the pattern is complex and there is enough data to learn it. Otherwise a simple model on good features usually does as well, and is easier to trust.

WHEN DO WE NEED COMPLEX MODELS?

WHEN DO WE NEED COMPLEX MODELS?

- *Data itself is highly complex (task like generating simulations)*
- *Pattern are complex; highly non-linear relationships; no clear relationship*
- *Useful features are unknown or difficult to find*
- *Summary statistics lose information*
- *Speed and volume*

Note: high dimensionality $\neq$ need for complex model

SPECIFIC OR FOUNDATION MODELS?

SPECIFIC OR FOUNDATION MODELS?

- *Distribution shift*
- *Selection effects inherited from training data*
- *Shortcut learning*
- *Uncalibrated uncertainty*
- *Interpretability*

So.....
pub?

